BANK MARKETING PREDICTION USING BOOSTING

Minor Project Report

Machine Learning Classification using Gradient Boosting

1. Abstract

This project develops a machine-learning workflow to predict whether a bank customer subscribes to a term deposit. The supplied bank-marketing CSV contains customer and campaign information, with **y** representing the subscription outcome. The notebook performs data inspection, data-quality checks, exploratory data analysis, preprocessing, model training, and evaluation using a Gradient Boosting Classifier. Final metric values and chart-specific findings must be added after running the notebook; they are not fabricated here.

2. Problem Statement

Banks contact customers to promote term-deposit products. This project investigates whether customer and campaign features can help predict subscription. The target variable is **y**: “yes” means subscribed and “no” means not subscribed. This is a binary classification problem.

3. Objectives

• Understand the dataset structure, columns, and target variable.

• Check missing values, duplicate records, and data types.

• Explore patterns using at least four meaningful visualizations.

• Prepare numerical and categorical features for machine learning.

• Train a Gradient Boosting Classifier and evaluate it with suitable metrics.

• Summarize findings, limitations, and possible improvements.

4. Dataset Description

Property	Description
Dataset	Bank Marketing (provided CSV file)
Initial size	4,521 rows × 17 columns
Target column	y
Target labels	yes / no
Task	Binary classification
Model	Gradient Boosting Classifier

The notebook loads the CSV using **sep=';'** because the fields are separated by semicolons. The **duration** feature is excluded because call duration is generally unknown before contacting a customer and could make the model less useful for pre-contact prediction.

5. Methodology

Data loading: Load the CSV into a pandas DataFrame with the correct delimiter.

Data understanding: Review dimensions, column names, data types, descriptive statistics, and target counts.

Data quality: Inspect missing values and duplicates, remove duplicate rows if present, and remove rows with missing target values.

Exploratory analysis: Visualize target distribution, age by target, balance by target, and subscription rate by education.

Preprocessing: Map the target to 0/1, exclude target and duration from the features, and one-hot encode categorical columns.

Train-test split: Use 80% training and 20% testing data with stratification and a fixed random seed.

Model training: Fit Gradient Boosting with 100 estimators, learning rate 0.05, maximum tree depth 2, and random_state 42.

Evaluation: Calculate accuracy, precision, recall, F1-score, ROC-AUC, and inspect the confusion matrix.

6. Visualizations and Observations

Visualization	Purpose
Target distribution	Compare yes/no counts; a large difference indicates class imbalance.
Age distribution	Explore subscription patterns across age ranges.
Account balance boxplot	Compare medians, spread, and extreme values between groups.
Education subscription rate	Compare subscription percentages by education; this does not prove causation.
Confusion matrix	Shows correct predictions and errors for each target class.
ROC curve	Shows ranking performance across classification thresholds.

After running the notebook, write one or two sentences describing the actual pattern seen in each chart.

7. Model Evaluation

Fill in the values below from the notebook output. They are left blank so the report does not claim results that have not been calculated.

Metric	Result	Meaning
Accuracy	_____	Overall proportion of correct predictions.
Precision	_____	How many predicted subscribers actually subscribed.
Recall	_____	How many actual subscribers were identified.
F1-score	_____	Balance between precision and recall.
ROC-AUC	_____	Ability to rank subscribers above non-subscribers.

Because the target may be imbalanced, accuracy should not be interpreted alone. Precision, recall, F1-score, the confusion matrix, and ROC-AUC provide additional information.

8. Conclusion

This project demonstrates an end-to-end workflow for bank-marketing subscription prediction. It includes data understanding, quality checks, visualization, categorical encoding, train-test splitting, Gradient Boosting training, and evaluation. Update the final conclusion with the actual chart observations and metric values after running the notebook.

9. Limitations and Future Work

• The dataset may represent a particular bank or period and may not generalize to all customers.

• Imbalanced classes can make accuracy misleading.

• Observed relationships do not establish causation.

• Future work could compare other classifiers, tune parameters, adjust the classification threshold, and validate on newer data.

10. Dataset Source and Tools

Dataset: user-provided CSV file. Confirm the original Kaggle page, dataset description, and license before submission, as required by the assignment. Tools: Python, pandas, NumPy, Matplotlib, Seaborn, and scikit-learn.